

# Notes on MML: Vectors, the Minus-One Trick, Linear Maps, Basis Change, Kernel, and Image

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## §1.1 Vectors beyond arrows

The first page stresses that vectors are not only geometric arrows. Polynomials can form a vector space: they can be added and scaled. Audio signals can also be treated as vectors: an audio signal is represented by a list of numbers, and adding or scaling those lists gives another signal.

The handwritten remark connects this to embeddings. In machine learning, word vectors or sentence vectors are not geometric arrows in the elementary sense, but they live in vector spaces and obey vector operations. Their meaning comes from the representation learned by the model, not from a literal physical arrow.

The lesson is that a vector is an element of a vector space. Geometry is only one model.

## §1.2 The minus-one trick

The source discusses a practical trick for reading solutions of a homogeneous system

$$Ax = 0, \quad A \in \mathbb{R}^{k \times n}.$$

Assume  $A$  is in reduced row echelon form and has no zero rows. Pivot columns correspond to leading ones; free columns correspond to variables that can be chosen freely.

The minus-one trick augments the row-reduced matrix into an  $n \times n$  matrix by inserting rows with a single  $-1$  in the diagonal positions corresponding to free variables. Then the columns containing  $-1$  on the diagonal directly give basis vectors for the null space.

The point is not magic. It is a compact way of doing the usual free-variable substitution. Setting one free variable to 1 and the others to 0 gives one kernel basis vector; the inserted  $-1$  row records that choice.

## §1.3 Example of the trick

The note's example starts with

$$A = \begin{bmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -4 \end{bmatrix}.$$

The free variables are in columns 2 and 5. Augmenting gives

$$\tilde{A} = \begin{bmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}.$$

The columns with diagonal  $-1$  give

$$x = \lambda_1 \begin{bmatrix} 3 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 3 \\ 0 \\ 9 \\ -4 \\ -1 \end{bmatrix}.$$

This is the same solution one obtains by solving by free variables, but the trick makes the basis easy to read.

## §1.4 Linear mappings

A mapping  $\Phi : V \rightarrow W$  between vector spaces is linear if

$$\Phi(\lambda x + \psi y) = \lambda \Phi(x) + \psi \Phi(y)$$

for all  $x, y \in V$  and scalars  $\lambda, \psi$ .

The note lists special cases:

- an isomorphism is linear and bijective;
- an endomorphism is a linear map  $V \rightarrow V$ ;
- an automorphism is a bijective endomorphism;
- the identity map  $\text{id}_V : V \rightarrow V$  is the identity automorphism.

The handwritten annotations translate these as “same structure,” “same domain and codomain,” and “self-isomorphism.”

## §1.5 Complex numbers as a real vector space

The example

$$\Phi : \mathbb{R}^2 \rightarrow \mathbb{C}, \quad \Phi(x_1, x_2) = x_1 + ix_2$$

is linear over  $\mathbb{R}$ :

$$\Phi(x + y) = \Phi(x) + \Phi(y), \quad \Phi(\lambda x) = \lambda \Phi(x).$$

It is also bijective, so it is an isomorphism of real vector spaces.

The note marks this as a homomorphism. The important distinction is that it preserves the vector-space addition and scalar multiplication. It is not claiming that every additional structure is automatically preserved.

## §1.6 Finite-dimensional isomorphism theorem

The theorem quoted from Axler says that finite-dimensional vector spaces  $V$  and  $W$  are isomorphic iff

$$\dim V = \dim W.$$

This is one of the most important simplifications in finite-dimensional linear algebra: after choosing bases, all  $n$ -dimensional vector spaces over the same field look like  $\mathbb{F}^n$ .

But this isomorphism is not canonical unless a basis is chosen. The abstract vector space and its coordinate representation are related but not identical.

## §1.7 Basis change

Let  $\Phi : V \rightarrow W$  be linear. Suppose  $V$  has old basis

$$B = (b_1, \dots, b_n)$$

and new basis

$$\tilde{B} = (\tilde{b}_1, \dots, \tilde{b}_n).$$

Similarly, let  $W$  have old basis

$$C = (c_1, \dots, c_m)$$

and new basis

$$\tilde{C} = (\tilde{c}_1, \dots, \tilde{c}_m).$$

If  $A_\Phi$  is the matrix of  $\Phi$  with respect to  $B, C$ , and  $\tilde{A}_\Phi$  is the matrix with respect to  $\tilde{B}, \tilde{C}$ , then

$$\tilde{A}_\Phi = T^{-1} A_\Phi S.$$

Here  $S$  converts coordinates from the new basis of  $V$  to the old basis of  $V$ , and  $T$  converts coordinates from the new basis of  $W$  to the old basis of  $W$ .

The diagram in the source page says exactly this: to use the old matrix, first translate the input coordinates into the old basis, then apply  $A_\Phi$ , then translate the output coordinates into the new output basis.

## §1.8 Equivalence and similarity

Two rectangular matrices  $A, \tilde{A} \in \mathbb{R}^{m \times n}$  are equivalent if there exist invertible matrices  $S \in \mathbb{R}^{n \times n}$  and  $T \in \mathbb{R}^{m \times m}$  such that

$$\tilde{A} = T^{-1} A S.$$

This corresponds to changing the basis in both domain and codomain.

Two square matrices  $A, \tilde{A} \in \mathbb{R}^{n \times n}$  are similar if

$$\tilde{A} = S^{-1} A S.$$

This corresponds to changing basis for a linear map  $V \rightarrow V$  using the same coordinate change in domain and codomain.

The handwritten table emphasizes the difference:

| Concept     | Situation                    | Invariant meaning                   |
|-------------|------------------------------|-------------------------------------|
| Equivalence | linear map $V \rightarrow W$ | change bases in domain and codomain |
| Similarity  | linear map $V \rightarrow V$ | same operator in a new basis        |

## §1.9 Kernel and image

For a linear map  $\Phi : V \rightarrow W$ , the kernel is

$$\ker \Phi = \{v \in V : \Phi(v) = 0_W\},$$

and the image is

$$\operatorname{Im} \Phi = \{w \in W : \exists v \in V, \Phi(v) = w\}.$$

The kernel records what is collapsed to zero. The image records what outputs are reachable.

The source diagram shows the domain on the left, codomain on the right, a yellow region for the kernel, and a blue/yellow region for the image. The note also records the injectivity criterion:

$$\Phi \text{ is injective} \iff \ker \Phi = \{0_V\}.$$

## §1.10 Categorical side remarks

The margin notes compare kernel, image, and cokernel with categorical language. In ordinary linear algebra,

$$\ker f \rightarrow A \rightarrow B$$

means the kernel is a subobject of the domain. The image is a subobject of the codomain. The cokernel, when defined, is a quotient-like object on the codomain side.

This is a useful preview: linear algebra already contains many category-theoretic patterns, even before they are named abstractly.

## §1.11 The general pattern

This note links computational linear algebra with structural thinking. Row reduction gives the null space; the minus-one trick packages free-variable substitution. Linear maps become matrices only after choosing bases. Basis change explains why the same map can have different matrices. Kernel and image describe what a map destroys and what it reaches. These ideas are also the language behind machine-learning embeddings, where vectors need not be arrows but still live in vector spaces with meaningful linear structure.

## §1.12 Vectors as elements of a linear structure

A vector is not essentially an arrow; it is an element of a vector space. Coordinates appear only after choosing a basis. The same vector has different coordinate columns in different bases.

## §1.13 The minus-one trick as a kernel graph

The phrase “minus-one trick” should be understood carefully. In this note it is not primarily a homogeneous-coordinate trick and it is not an affine shift. It is a compact way of reading the kernel of a row-reduced homogeneous system.

**Proposition 1.13.1** (Kernel graph form of an RREF system)

After possibly reordering variables, suppose a homogeneous system has row-reduced matrix

$$\begin{bmatrix} I_r & A \end{bmatrix}.$$

Write the unknown vector as

$$x = (x_p, x_f),$$

where  $x_p \in k^r$  are pivot variables and  $x_f \in k^{n-r}$  are free variables. Then the solution space is

$$\ker \begin{bmatrix} I_r & A \end{bmatrix} = \left\{ \begin{pmatrix} -Au \\ u \end{pmatrix} : u \in k^{n-r} \right\}.$$

Thus the kernel is the graph of the linear map

$$u \mapsto -Au.$$

*Proof.* The equation is

$$x_p + Ax_f = 0.$$

Hence  $x_p = -Ax_f$ . If we write  $u = x_f$ , every solution has the displayed form, and every displayed vector clearly satisfies the equation.  $\square$

The inserted  $-1$  rows in the practical trick are a bookkeeping device for the free variables. Choosing  $u = e_j$ , the  $j$ -th standard basis vector of the free-variable space, gives one kernel basis vector:

$$\begin{pmatrix} -Ae_j \\ e_j \end{pmatrix}.$$

So the trick is best read as a way of writing the graph basis of a kernel.

**Remark 1.13.2** (Do not confuse two uses of extra coordinates) — Homogeneous coordinates do turn affine transformations into linear transformations one dimension higher. That is a real and useful idea. But the minus-one trick in this row-reduction context is different: it does not linearize an affine map; it packages the free-variable choices in a homogeneous linear system.

## §1.14 Kernel-image factorization

For a linear map  $T : V \rightarrow W$ , the kernel records lost directions and the image records reachable directions. Rank-nullity says

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

## §1.15 Coordinates, kernels, and affine linearization

The review should distinguish vectors from coordinates, affine maps from linear maps, and a matrix from the linear transformation it represents. Kernel, image, and basis change are the structural core.

## §1.16 Exact sequences behind rank-nullity

For a linear map  $T : V \rightarrow W$ , the kernel and image fit into the exact sequence

$$0 \rightarrow \ker T \rightarrow V \xrightarrow{T} \operatorname{im} T \rightarrow 0.$$

Exactness means that the image of each arrow equals the kernel of the next. Taking dimensions gives

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

This is rank-nullity.

The elementary theorem is therefore the dimension shadow of an exact sequence. Exact sequences become one of the central languages of algebra, topology, and geometry.

## §1.17 Quotient description of the image

The first isomorphism theorem says

$$V / \ker T \cong \operatorname{im} T.$$

The quotient collapses all vectors that differ by a null vector. What remains is precisely the information visible after applying  $T$ .

This gives a clean meaning to free variables and solution spaces. The kernel is what the map forgets; the image is what survives.

## §1.18 Cokernel and obstructions

The cokernel of  $T : V \rightarrow W$  is

$$\operatorname{coker} T = W / \operatorname{im} T.$$

It measures the failure of  $T$  to be onto. A system

$$Tx = b$$

is solvable exactly when  $b$  maps to zero in the cokernel.

Thus kernel measures non-uniqueness, while cokernel measures obstruction to existence. This is the conceptual form of the elementary distinction between infinitely many solutions and no solution.

## §1.19 Kernel, image, and quotient survive basis changes

The minus-one trick, basis conversion, and kernel/image computations in the note are all examples of one structure: a linear map forgets some directions, preserves some information, and may fail to reach some outputs. Exact sequences and quotients express this without depending on a chosen matrix.

## §1.20 Row reduction as choosing a splitting

When a matrix is row-reduced, the pivot variables are expressed in terms of free variables. Abstractly, this chooses a splitting of the domain into a complement of the kernel plus the kernel:

$$V \cong U \oplus \ker T.$$

The map  $T$  is injective on the chosen complement  $U$  and maps  $U$  isomorphically onto  $\operatorname{im} T$ .

This splitting is not canonical: different row-reduction choices or different bases may choose different complements. The exact sequence

$$0 \rightarrow \ker T \rightarrow V \rightarrow \operatorname{im} T \rightarrow 0$$

is canonical; a particular parametric solution is a coordinate choice.

## §1.21 Cokernel as the missing equation space

For  $T : V \rightarrow W$ , the cokernel  $W / \operatorname{im} T$  measures which right-hand sides are not reachable. In matrix terms, left null vectors produce compatibility conditions. If  $y \in \ker T^T$ , then any solvable equation  $Tx = b$  must satisfy

$$y^T b = 0.$$

Thus the left null space is the dual of the obstruction space.

This gives a precise interpretation of consistency conditions in linear systems: they are not extra tricks, but equations defining whether  $b$  vanishes in the cokernel.